



ABHAY KUMAR SINGH

Manufacturing Excellence & Digital Transformation Professional | Operational Excellence | Lean Six Sigma Black Belt

🔧 OEE Excellence

🛡️ Safety First

💰 Cost Optimization

🚚 Delivery Performance

📊 COPQ Reduction



Scan for LinkedIn

📞 +91-9504471475

✉️ profesionalabhay@gmail.com

📍 Chennai, India

🌐 linkedin.com/in/abhay-singh-Strategic-leader

PROFESSIONAL SUMMARY

Manufacturing Excellence & Digital Transformation Professional with 16+ years of experience advancing production performance, automation, TPM, and operational excellence across automotive and engineering environments. Lean Six Sigma Black Belt with proven success deploying smart manufacturing, SAP/MES integration, robotics, Power BI analytics, and continuous improvement systems to improve OEE, quality, cost, delivery, and safety. Experience leading shop-floor digitalization, manufacturing intelligence systems, and data-driven operational excellence initiatives

STRATEGIC FOCUS AREAS & IMPACT

🔧 OEE Excellence

Achieved 20% OEE improvement through structured loss elimination, real-time monitoring, and targeted automation across 15+ production lines

📊 COPQ Reduction

Reduced Cost of Poor Quality by minimizing defects (25% rework reduction), implementing SPC, and driving zero-defect culture

🛡️ Safety & Compliance

Implemented 5S, TPM safety pillars, and visual management systems ensuring 100% compliance with safety standards and zero major incidents

🚚 On-Time Delivery

Enhanced delivery performance through takt-based planning, WIP control, and MES integration improving planning accuracy by 20%

📊 Digital Manufacturing & AI Initiatives

- Developed AI-powered COPQ Review Deck Generator reducing manual reporting effort by over 90%.
- Leading development of Manufacturing Execution System (MES) for production tracking, OEE/OLE monitoring, WIP visibility, loss analysis and traceability.
- Developed automated analytics and dashboard solutions using Power BI, Excel VBA and AI-assisted workflows.
- Driving Industry 4.0 initiatives involving automation, robotics, digital manufacturing and data-driven decision making.

KEY ACHIEVEMENTS DASHBOARD

20%

OEE Improvement

₹23L+

Annual Savings

25%

COPQ Reduction

25%

Downtime Reduction

100%

Safety Compliance

20%

Delivery Performance

100+

Professionals Trained

15%

Cycle Time Reduction

CORE COMPETENCIES

Operations & Production Excellence

- Production Planning & Execution
- Line Balancing & Takt Time Optimization
- Capacity Planning & WIP Control
- Process Standardization & Daily Work Management
- Shop Floor Performance Management
- Operator Productivity & Shift Governance

Lean & Continuous Improvement

- Robotics, PLCs & Industrial Automation
- TPM Deployment & Predictive Maintenance
- Lean Six Sigma Black Belt
- Kaizen, 5S, SMED & VSM
- OEE Improvement & Loss Elimination
- Root Cause Analytics (4M/8D/5 Why)
- COPQ Reduction & Defect Prevention

Data Analytics & Technology

- CAPA & Corrective Action Governance
- Process Capability & Defect Prevention
- Quality Data Analysis & Visual Management
- Audit Readiness & Control Discipline
- Operational Risk Reduction

Digital Manufacturing & Transformation

Quality, Safety & Governance

Leadership & People Development

- Smart Manufacturing & Industry 4.0
- Manufacturing Analytics & Digital KPI Systems
- SAP/MES Integration & Shop Floor Digitalization
- Real-Time Production Monitoring
- Data Visualization & Manufacturing Intelligence

- Zero-Defect Culture & Quality Assurance
- Safety Systems & Compliance
- Cost of Poor Quality (COPQ) Management
- Statistical Process Control (SPC)
- FMEA & Risk Assessment

- Cross-Functional Team Leadership
- Capability Building & Skill Matrix Design
- Training Systems & Knowledge Transfer
- Change Leadership & Stakeholder Alignment
- Culture & Transformation

PROFESSIONAL EXPERIENCE

Nash India Industries Ltd | Chennai

May 2025 - Present

C.I Lead – Manufacturing Excellence

- Driving enterprise-wide Manufacturing Excellence by integrating Lean, TPM, Six Sigma, and digital performance management across production and support functions.
- Institutionalized 5Q-based Daily Work Management, visual takt boards, and escalation routines, improving issue-closure speed by 20%
- Built a Digital Loss Intelligence Platform integrated with SAP, reducing downtime analysis cycle time by 25%
- Reduced COPQ by 25% through defect analytics, SPC discipline, and zero-defect improvement campaigns
- Optimized takt-based manpower deployment, unlocking 250 operator hours/month while supporting delivery performance
- Developed real-time Power BI manufacturing intelligence dashboards for OEE, Safety, Cost, Delivery, and Quality, accelerating decisions by 40%
- Expanded skill-matrix coverage by 20% and reduced operator ramp-up time by 30% through structured capability development
- Strengthened TPM-led safety systems, autonomous maintenance, and near-miss governance to sustain 100% safety compliance

KEY METRICS: 20% OEE improvement | 25% COPQ reduction | 250 operator hrs/month saved | 40% faster decisions | 100% safety compliance

Steel Strips Wheels Ltd | Chennai & Jamshedpur

Apr 2015 - Apr 2025 (10 Years)

Senior Engineer - Production & Automation

- Led press shop and robotic weld shop operations for major automotive OEM programs, balancing throughput, quality, and automation performance
- Led cross-functional teams of 100+ operators, supervisors, and engineers across 15+ production lines.
- Delivered 15% takt time reduction through line balancing, process optimization, and automation enhancement
- Engineered Fanuc auto-fixture logic, increasing production throughput by 12%
- Implemented MES-enabled WIP visibility and digital dashboards, improving production planning accuracy by 20%
- Led 8D and 4M root-cause analytics, reducing weld rework by 25% and saving ₹8 lakh/year in scrap cost
- Deployed predictive maintenance analytics and maintenance planning, improving equipment uptime by 20%

KEY METRICS: 15% takt time reduction | 12% throughput increase | 20% uptime improvement | ₹8 lakh annual scrap savings

New Vishal Engineering | Jamshedpur

Dec 2012 - Mar 2015

Electrical Maintenance Engineer

- Reduced critical machine failures by 30% through preventive maintenance discipline and breakdown analytics
- Executed electrical panel modernization and energy audits, improving operating efficiency by 12%
- Standardized electrical troubleshooting protocols, cutting downtime by 18%
- Developed PM schedules and maintenance tracking systems for 50+ machines, strengthening reliability control

KEY METRICS: 30% failure reduction | 18% downtime reduction | 12% energy savings

- Delivered 20+ multi-site telecom infrastructure rollouts on time and within budget
- Coordinated 12+ vendors and contractors while maintaining 100% quality and safety compliance
- Managed cross-functional execution, scheduling, and stakeholder coordination across concurrent sites

KEY METRICS: 30% failure reduction | 18% downtime reduction | 12% energy savings

KEY PROJECTS & INITIATIVES

5Q Daily Work Management System

Structured governance model improving issue-closure speed by 20%

Shop Floor Digital Performance Management System

Visual takt and accountability framework reducing process delays by 15%

Digital Loss Intelligence Platform

SAP/MES-linked downtime intelligence system reducing response time by 25%

Digital Operator Enablement App

Mobile-first workflow concept reducing shop-floor data-entry errors by 35%

Real-Time Manufacturing Intelligence Dashboard

Power BI KPI cockpit accelerating management decision-making by 40%

Robotic Fixture Automation Logic

Automated robot programming sequence saving 200 hours/year

Predictive Maintenance Analytics System

Condition-based monitoring model delivering ₹15 lakh/year in avoided loss

Scrap Cost Reduction Analytics Initiative

4M/8D-driven defect analytics delivering ₹8 lakh annual savings

SKILLS

ERP & Manufacturing Systems

SAP PP/MM, Manufacturing Execution Systems (MES), Production Planning Systems, Shop Floor Reporting Platforms

Data Analytics & Visualization

Power BI, Advanced Excel, Excel VBA/Macros, SQL, KPI Dashboards, SPC Analytics

Automation & Robotics

Fanuc Robot Programming, Yaskawa Systems, Lincoln Welding

Systems, Seam Tracking, PLC Troubleshooting

Digital Manufacturing & Industry 4.0

Smart Manufacturing, Real-Time Production Monitoring, Digital KPI Systems, Manufacturing Intelligence, Predictive Maintenance Analytics

Programming & Analytics

Python, C, C++, Embedded C, SQL, Minitab, Control Charts, Pareto Analysis, FMEA, Cp/Cpk

EDUCATION & CERTIFICATIONS

EDUCATION

Bachelor of Technology (B.Tech)

Electronics & Telecommunication Engineering | 2010

PROFESSIONAL CERTIFICATIONS

✓ Lean Six Sigma Black Belt

SixSigmaOnline (Educate 360) | 2025

✓ Lean Six Sigma Green Belt

Aveta Business Institute | 2019

✓ Advanced Excel Certification

2023

languages:

English (Fluent), Hindi (Native)

Location Flexibility:

Open to pan-India

Availability:

30 to 90 days' notice